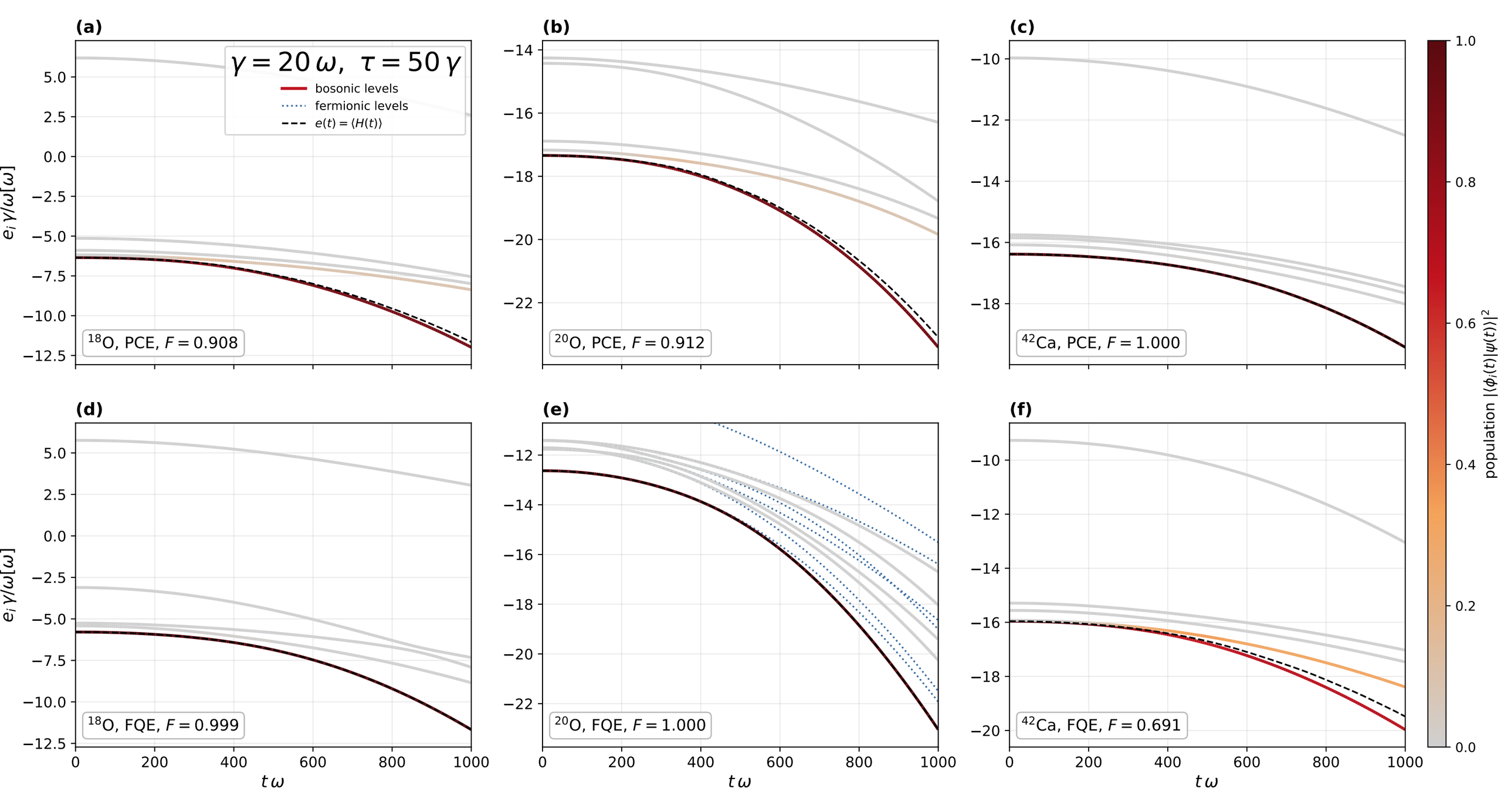

(a)
(b)
(c)
(d)
(e)
(f)
$\gamma = 20\,\omega,\ \tau = 50\,\gamma$
bosonic levels
fermionic levels
$e(t) = \langle H(t) \rangle$
^{18}O, PCE, $F = 0.908$
^{20}O, PCE, $F = 0.912$
^{42}Ca, PCE, $F = 1.000$
^{18}O, FQE, $F = 0.999$
^{20}O, FQE, $F = 1.000$
^{42}Ca, FQE, $F = 0.691$
$e_i\,\gamma/\omega[\omega]$
$t\,\omega$
population $|\langle\phi_i(t)|\psi(t)\rangle|^2$